

Discussion Assignment: What Makes Netflix's Recommender System Work (and What Should It Think About)

Goal

Understand what technical, business, and ethical factors a streaming platform's recommender system must consider to serve users well, stay competitive, and do so responsibly.

Instructions

1. Read the assigned article about Netflix's recommender system: [Netflix's Billion Dollar Secret](#).
2. In small groups, discuss the questions below. Take notes — you'll come back to share with the class.
3. Each group will present one or two insights they found most interesting (or surprising), as well as one challenge they foresee for Netflix or similar companies.

Category	Questions
User Experience & Engagement	• How should Netflix balance “surprise/novelty” vs. recommending what users are most likely to click/watch? • How should it handle the “cold start” problem (i.e., new users or new content)?
Diversity & Filter Bubbles	• How “narrow” or “broad” should recommendations be? Should Netflix try to expose users to genres/styles outside their history? • How might personalization lead to echo filter bubbles? • What strategies could mitigate this (e.g. mixing in less popular content)?
Business & Economics	• How do business objectives shape what the recommender should optimize for? (e.g., retention, watch time, content licensing costs, content production) • How might content procurement/licensing influence recommendation choices? • Should Netflix prefer to promote its own productions (“originals”)? What are the trade-offs?
Technical / ML Considerations	• What signals does Netflix likely use (e.g. watch completion, browsing behavior, time spent, rewatches)? How do those differ in information about consumer preferences? • How should Netflix test its recommendations? First on past data (hide last month and check whether the recommendations match

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	what people actually watched), then with real customers (an A/B test: half get the new recommendations, half the old). What should it measure in each test?
Fairness, Ethics & Privacy	• How to guard user privacy while collecting rich behavioral data? • How to ensure fairness (e.g. of content producers, minority genres, global vs local content)? • Could recommendation systems reinforce stereotypes or biases (e.g. by over-promoting certain types of content)?
Long-Term vs Short-Term Goals	• Should Netflix optimize for immediate watch time / clicks, or for long-term satisfaction / loyalty? • How might “over-optimization” for clicks degrade long-term content quality or user experience?